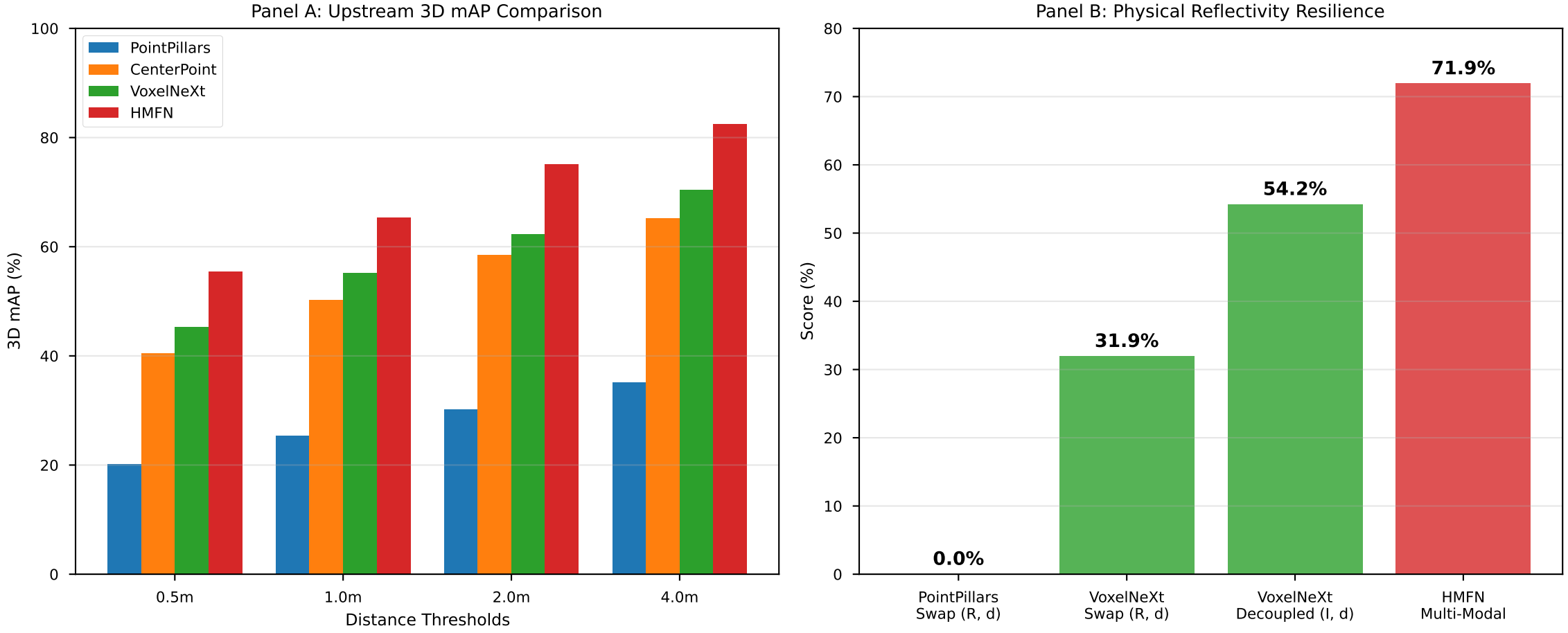
Panel A: Upstream 3D mAP Comparison
PointPillars
CenterPoint
VoxelNeXt
HMFN
3D mAP (%)
0
20
40
60
80
100
0.5m
1.0m
2.0m
4.0m
Distance Thresholds
Panel B: Physical Reflectivity Resilience
Score (%)
0
10
20
30
40
50
60
70
80
0.0%
31.9%
54.2%
71.9%
PointPillars
Swap (R, d)
VoxelNeXt
Swap (R, d)
VoxelNeXt
Decoupled (I, d)
HMFN
Multi-Modal